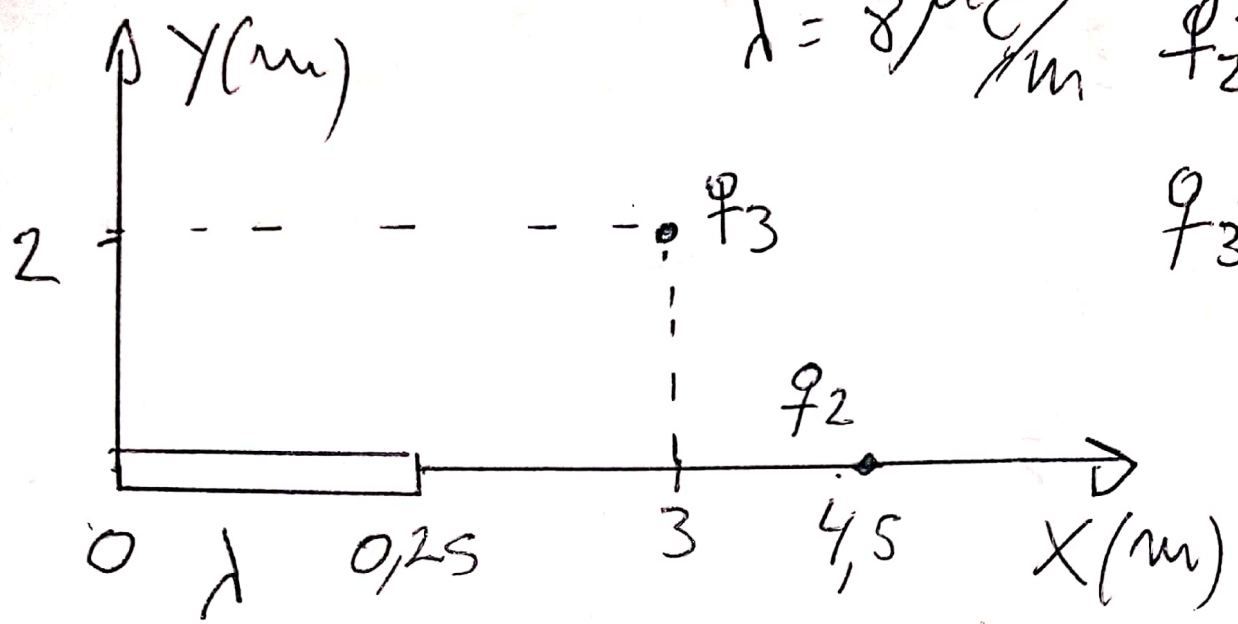
EJERCICIO 6:



$\lambda = 8\,\mu C/m$ $q_2 = -3\,\mu C$

$q_3 = 1\,\mu C$

a) CARGA DE LA VARILLA:

$$Q = \int \lambda\, dl = \lambda \int dl = \lambda\, l = 8\times10^{-6}\,\frac{C}{m} \cdot 0{,}25\,m$$

HILO $dl = dx$, $\lambda = CTE.$

$Q = 2\,\mu C$

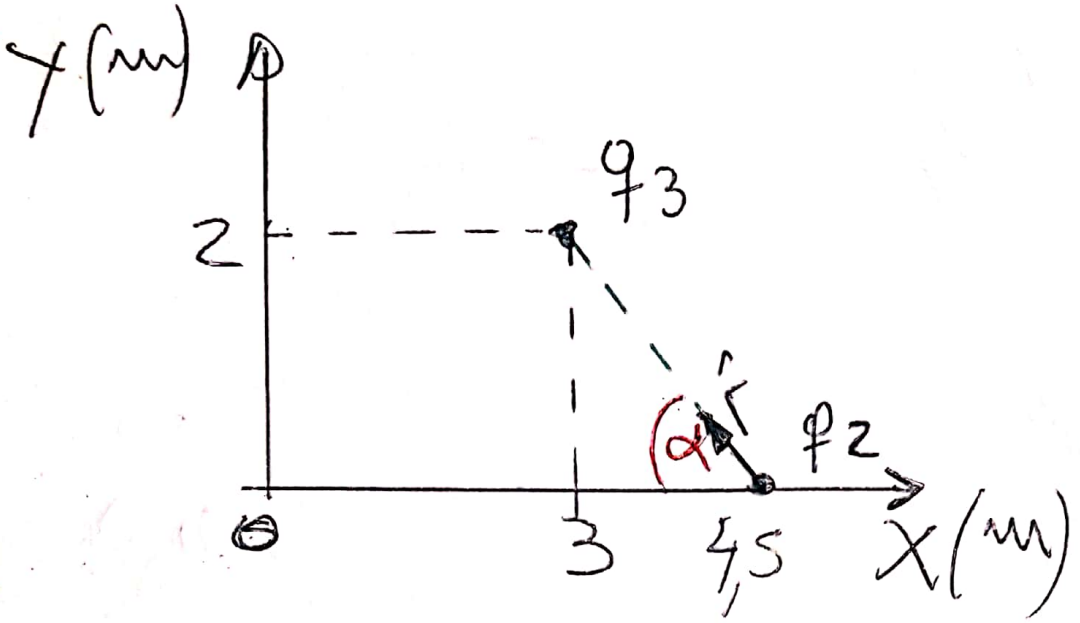
((COINCIDE CON q_1 DEL EJ. 5))

b) FUERZA SOBRE q_3 DEBIDA AL HILO Y A q_2

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EJERCICIO 6 (CONT.)

COMENCEMOS CON LA FUERZA DEBIDA A q_2:



$$\operatorname{tg}\alpha = \frac{2}{4,5-3} = \frac{2}{1,5} \rightarrow \alpha = 53,13^\circ$$

$$\vec{F}_{3,2} = q_3 \vec{E}_{q_2} = q_3 \frac{k q_2}{r^2} \hat{r} = k \frac{q_2 q_3}{r^2} \hat{r}$$

$$\hat{r} = -\cos\alpha\,\hat{i} + \operatorname{sen}\alpha\,\hat{j} = -0,6\,\hat{i} + 0,8\,\hat{j}$$

$$r^2 = (2\,m)^2 + (4,5\,m - 3\,m)^2 \rightarrow r^2 = 6,25\,m^2$$

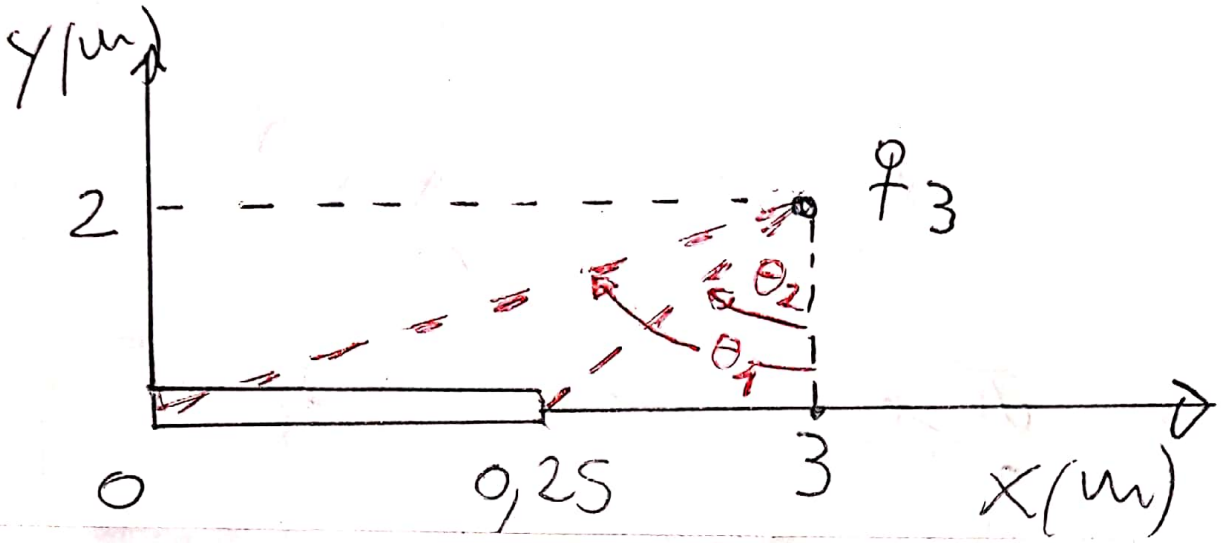
$$\vec{F}_{3,2} = 9\times10^9 \frac{N\,m^2}{C^2} \times \frac{(-3)\times10^{-6}\,C \times 1\times10^{-6}\,C}{6,25\,m^2} \times (-0,6\,\hat{i} + 0,8\,\hat{j})$$

$$\vec{F}_{3,2} = (2,59\,\hat{i} - 3,46\,\hat{j})\times10^{-3}\,N$$

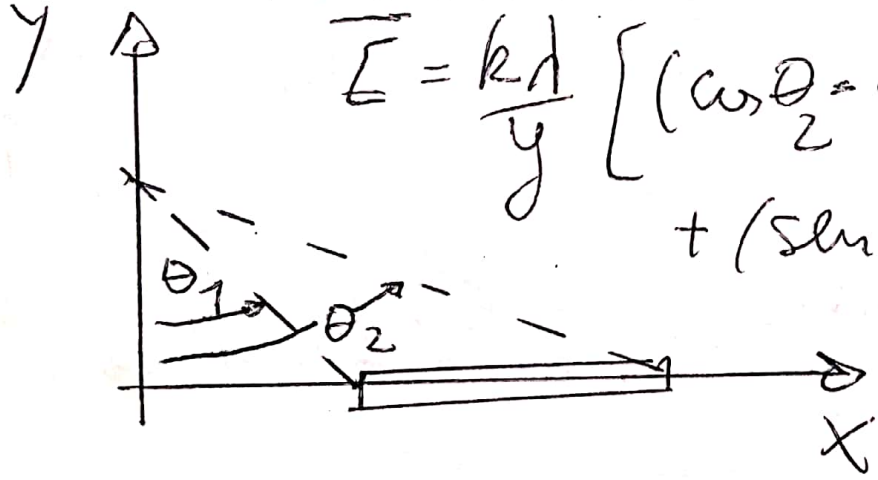
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EJERCICIO 6 (CONT.)

AHORA VEAMOS EL HILO:



RECORDEMOS:



$$\vec{E} = \frac{k\lambda}{y}\left[(\cos\theta_2 - \cos\theta_1)\,\hat{i} + (\operatorname{sen}\theta_2 - \operatorname{sen}\theta_1)\,\hat{j}\right]$$

EN NUESTRO CASO $\theta_1 < 0$, $\theta_2 < 0$

$$\operatorname{tg}\theta_1 = -\frac{3}{2} \rightarrow \theta_1 = -56,31°$$

$$\operatorname{tg}\theta_2 = -\frac{3-0,25}{2} = -\frac{2,75}{2} \rightarrow \theta_2 = -53,97°$$

$y = 2\,m$ REEMPLAZANDO:

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EJERCICIO 6 (CONT.)

$$\vec{F}_{3,HILO} = q_3 \vec{E}_{HILO} =$$ ¡OJO CON LOS SIGNOS!

$$= 1\times10^{-6}\,C \cdot 9\times10^{9}\,\frac{N m^2}{C^2} \times 8\times10^{-6}\,\frac{C}{m} \times \frac{1}{2m} \times$$

$$\times\left[\left(\cos(53{,}97) - \cos(56{,}31)\right)\hat{\imath} - \left(\operatorname{sen}(53{,}97) - \operatorname{sen}(56{,}31)\right)\hat{\jmath}\right] =$$

$$= 0{,}036\,N\left(0{,}0335\,\hat{\imath} + 0{,}0233\,\hat{\jmath}\right)$$

$$\vec{F}_{3,HILO} = \left(1{,}21\,\hat{\imath} + 0{,}84\,\hat{\jmath}\right)\times10^{-3}\,N$$

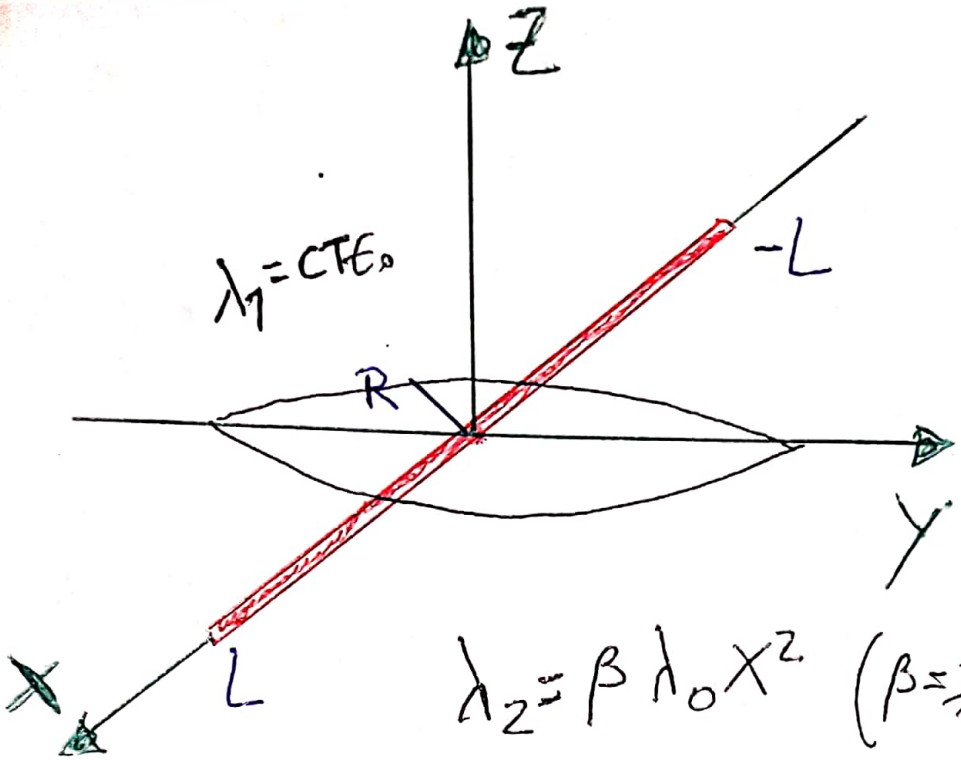
LUEGO $\vec{F}_3 = \vec{F}_{3,2} + \vec{F}_{3,HILO}$

$$\boxed{\vec{F}_3 = \left(3{,}8\,\hat{\imath} - 2{,}62\,\hat{\jmath}\right)\times10^{-3}\,N}$$

c) EXPLICADO EN LA RESPUESTA

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EJERCICIO 13



$\lambda_2 = \beta \lambda_0 x^2 \quad \left(\beta = \frac{1}{m^2}\right)$

$$Q_{ANILLO} = \lambda_1 2\pi R$$

$$Q_{HILO} = \int_{HILO} \delta\varphi = \int_{HILO} \lambda_2 dl = \int_{-L}^{L} \lambda_2 dx = \beta\lambda_0 \int_{-L}^{L} x^2 dx \qquad (dl = dx)$$

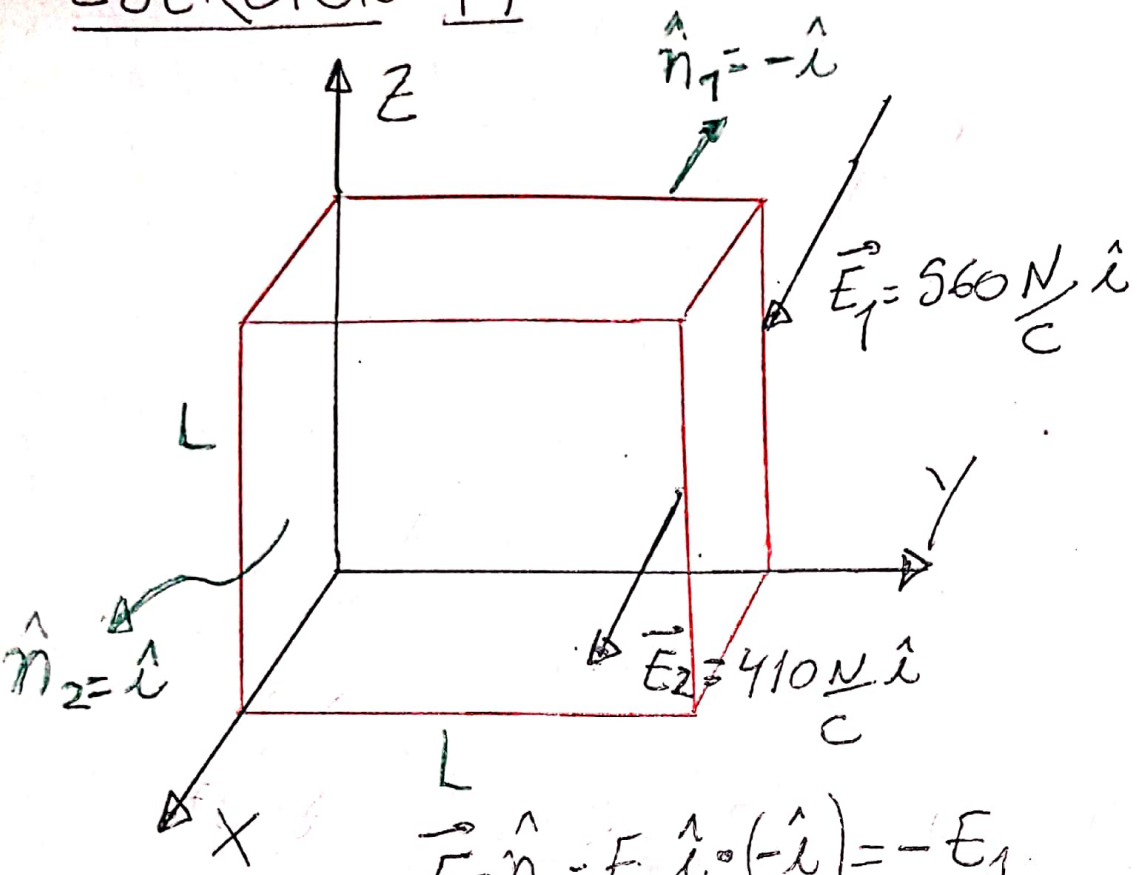
$$Q_{HILO} = \beta \frac{\lambda_0}{3}\left[L^3 - (-L)^3\right] \longrightarrow Q_{HILO} = \frac{2}{3}\beta\lambda_0 L^3$$

$$0 = Q_{ANILLO} + Q_{HILO} \longrightarrow \frac{2}{3}\beta\lambda_0 L^3 = -\lambda_1 2\pi R \longrightarrow$$

$$\longrightarrow \boxed{\lambda_0 = -3\lambda_1 \frac{R}{\beta L^3}\pi}$$

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EJERCICIO 19



$$\vec{E}_1 \cdot \hat{n}_1 = E_1 \hat{\imath} \cdot (-\hat{\imath}) = -E_1$$

$$\vec{E}_2 \cdot \hat{n}_2 = E_2 \hat{\imath} \cdot \hat{\imath} = E_2$$

SOLO SE GENERA FLUJO A TRAVÉS DE LAS DOS CARAS PARALELAS AL PLANO Y-Z

$$\frac{Q_{ENC}}{\varepsilon_0} = \phi = \left(\vec{E}_1 \cdot \hat{n}_1 + \vec{E}_2 \cdot \hat{n}_2\right) L^2 = \left(-E_1 + E_2\right) L^2$$

$$Q_{ENC} = 8{,}85 \times 10^{-12} \frac{C^2}{Nm^2} \times \left(-560 \frac{N}{C} + 410 \frac{N}{C}\right) \times (25\,m)^2$$

$$\boxed{Q_{ENC} = -0{,}983\,\mu C}$$

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